# PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

October 23, 2026

# Evolving into Agents



"Airlines from Harrisburg to Orlando"



"What is the best time to leave State College for a Harrisburg flight?"



"Book me a flight from Harrisburg to Orlando on November 20."

# What is an Agent?

**"An AI agent is a language model running in a loop with tools."**

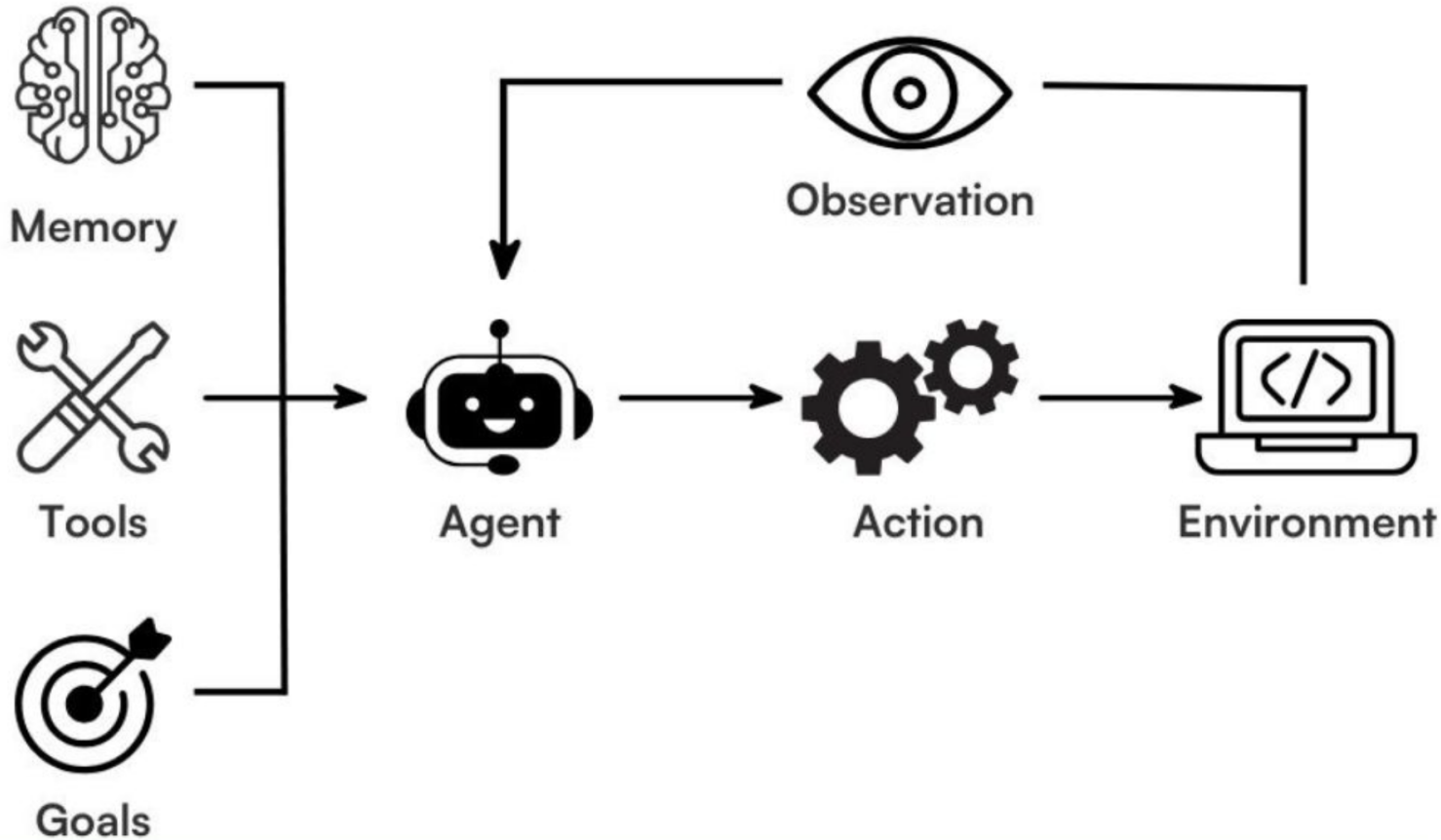
Four parts:

- A Goal
- Tools
- A Loop
- Autonomy

Mouschoutzi:

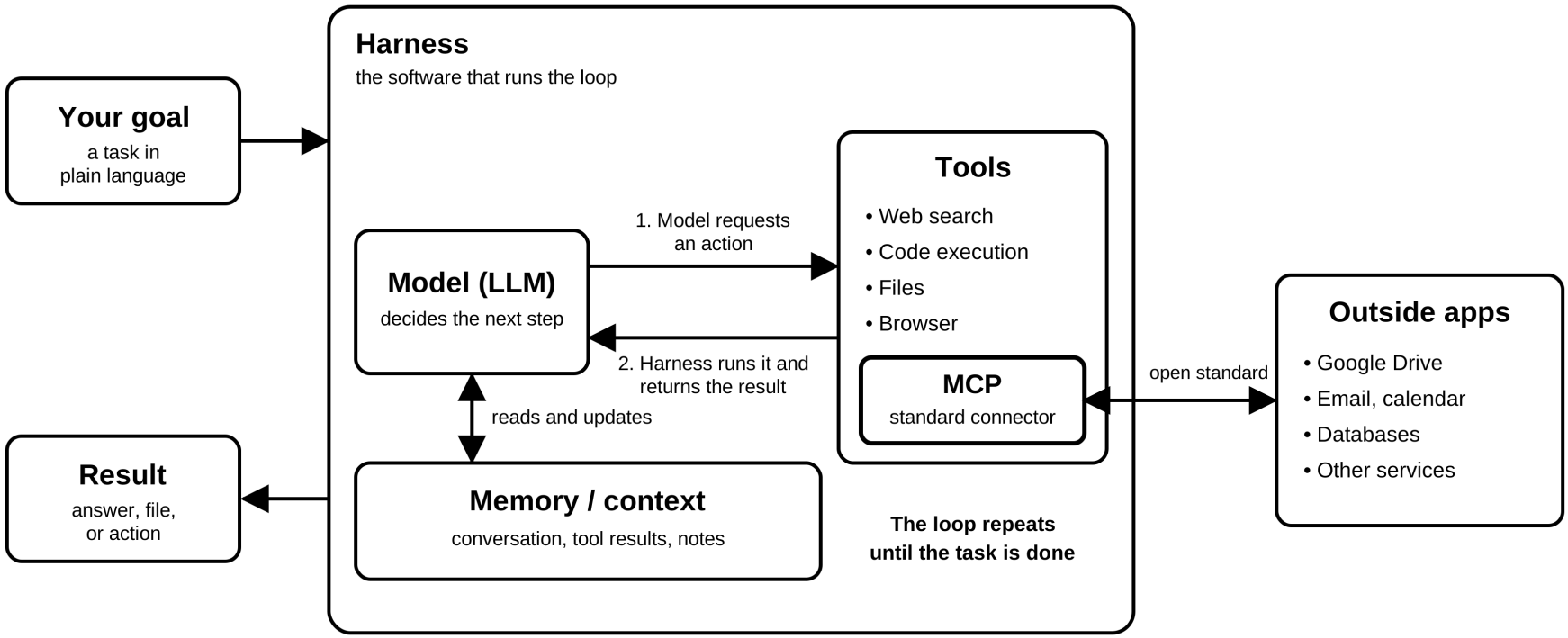
**Perceive → Decide → Act**

# Agentic AI Explained




(source)

# Anatomy of an Agent



# Example: Fact-Checking

**“A friend sent me a social media post claiming that 88 percent of Americans believe that marijuana should be legal for recreational use. Is this accurate? Find where the number comes from, check it against the original data, and tell me what holds up and what doesn’t.”**

# Fact-Checking, continued

| Step | Loop Phase | What the Agent Does | Components |
|---|---|---|---|
| 1 | Think | Breaks the goal into subtasks: find the origin, find the primary data, compare the numbers, report | Model, Memory |
| 2 | Act | Searches for the claim's origin | Tools (web search) |
| 3 | Observe, Think | Finds a news article citing "a recent survey," with no link | Memory |
| 4 | Act | Follows the article's link and gets a dead page | Tools (browser) |
| 5 | Think, Act | Changes approach: searches the survey organization's site and an archived copy for the report | Model, Tools |
| 6 | Act, Observe | Downloads the report PDF and extracts the table with code. The post's number appears, but for a different question wording than the post implies | Tools (files, code) |
| 7 | Act | Optionally recomputes from the published toplines or data to confirm | Tools (code) |
| 8 | Stop, report | Gives a verdict, links, and caveats, then offers to draft a reply to your friend and *does not send it* | Harness |

# What Can Go Wrong...

| Failure Mode | Locus | Sources & Symptoms |
|---|---|---|
| Compounding Errors | (whole system) | "Silent failure" – but note that many systems are self-correcting |
| Prompt Injection | Tools $\rightarrow$ Model | Security breaches; worst w/agents combining access to private data + exposure to untrusted content + external communication |
| Loss of Control | MCP + Apps | Agent acts independently in unauthorized ways; solutions = "least privilege," read-only, human approval for outward-facing actions, detailed logs |
| Accountability | (whole system) | Autonomous agents leave open the question of responsibility when things go wrong |

# Why This Matters

Applications, etc.:

- Agents as Research Assistants (Mollick, etc.)
- Agents in the Information Environment (e.g., the UZ Reddit experiment)
- Agents as Subjects (Argyle et al. 2023 vs. Bisbee et al. 2024)
- Governance and Liability (ethics, legal liability, disclosure, etc.)